1 Ohmscher Widerstand

U [V]	I [A]	σ_U [V]	R [Ω]	σ_R [Ω]
0.943	0.002	0.01	471.5	6.9
1.87	0.004	0.01	467.5	3.4
2.828	0.006	0.01	471.3	2.3
3.769	0.008	0.01	471.1	1.7
4.68	0.01	0.01	468	1.4
5.627	0.012	0.01	468.9	1.1
6.581	0.014	0.01	470.1	1
7.517	0.016	0.01	469.8	0.9
8.46	0.018	0.01	470	0.8
9.391	0.02	0.01	469.6	0.7

Table 1: widerstand

Calculations: $\sigma_U = \sigma_U$ (with $\sigma_U = 0.01\,\mathrm{V}$) $\quad R = \frac{U}{I} \quad$ $\sigma_R = \sqrt{\frac{\sigma_U^2}{I^2} + \frac{U^2 \cdot \sigma_I^2}{I^4}}$ (with $\sigma_I = 2 \times 10^{-5}\,\mathrm{A}$)

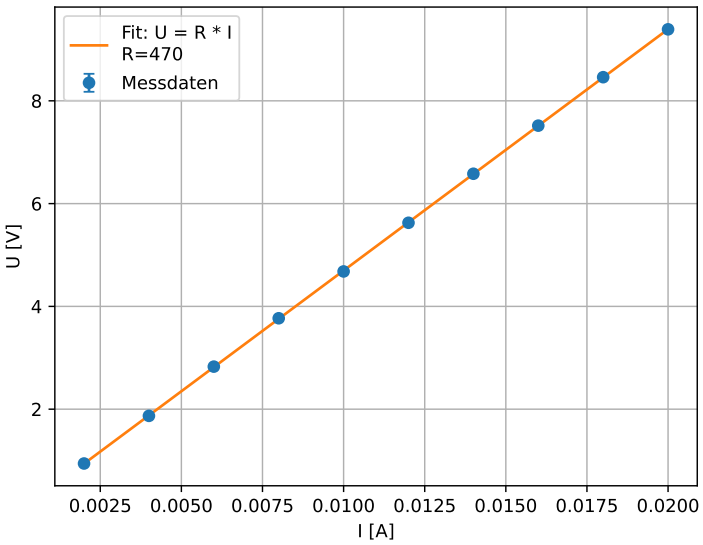


Figure 1: U [V] vs. I [A]

Regression: $U = I \cdot R$

parameter	value	error
R	469.69	± 0.25

$\chi^2/\text{ndf} = 7.62/9 = 0.847$

Calculations: $\sigma_U = \sigma_U$ (with $\sigma_U = 0.01\,\text{V}$)

2 RC-Tiefpass

f [Hz]	U_{in} [V]	U_{out} [V]	$\sigma_{U_{out}}$ [V]	H	σ_H
100	2	2.005	0.01	1.0025	0.0071
200	2	1.992	0.01	0.996	0.0071
400	2	1.94	0.01	0.97	0.007
700	2	1.842	0.01	0.921	0.0068
1000	2	1.698	0.01	0.849	0.0066
1500	2	1.447	0.01	0.7235	0.0062
2000	2	1.249	0.01	0.6245	0.0059
3000	2	0.928	0.01	0.464	0.0055
5000	2	0.615	0.01	0.3075	0.0052
8000	2	0.39	0.01	0.195	0.0051
12000	2	0.261	0.01	0.1305	0.005

Table 2: tiefpass

Calculations: $H = \frac{U_{out}}{U_{in}}$ $\sigma_H = \sqrt{\frac{\sigma^2_{U_{out}}}{U^2_{in}} + \frac{U^2_{out} \cdot \sigma^2_{U_{in}}}{U^4_{in}}}$ (with $\sigma_{U_{in}} = 0.01\,\mathrm{V}$)

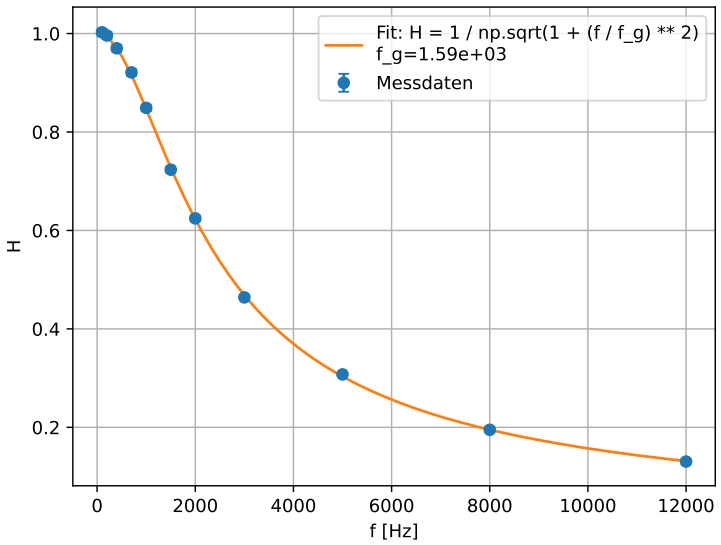


Figure 2: H vs. f [Hz]

Regression: $H = \frac{1}{\sqrt{\frac{f^2}{f_g^2}+1}}$

parameter	value	error
f_g	1591	± 12

$\chi^2/\text{ndf} = 3.45/10 = 0.345$

Calculations: $H = \frac{U_{out}}{U_{in}}$ $\qquad \sigma_H = \sqrt{\frac{\sigma^2_{U_{out}}}{U^2_{in}} + \frac{U^2_{out} \cdot \sigma^2_{U_{in}}}{U^4_{in}}}$ (with $\sigma_{U_{in}} = 0.01\,\text{V}$)

3 Rohdaten

U [V]	I [mA]
0.943	2.0
1.870	4.0
2.828	6.0
3.769	8.0
4.680	10.0
5.627	12.0
6.581	14.0
7.517	16.0
8.460	18.0
9.391	20.0

Table 3: widerstand (raw data)

f [Hz]	U_{in} [V]	U_{out} [V]	$\sigma_{U_{out}}$ [V]
100	2.00	2.005	0.010
200	2.00	1.992	0.010
400	2.00	1.940	0.010
700	2.00	1.842	0.010
1000	2.00	1.698	0.010
1500	2.00	1.447	0.010
2000	2.00	1.249	0.010
3000	2.00	0.928	0.010
5000	2.00	0.615	0.010
8000	2.00	0.390	0.010
12000	2.00	0.261	0.010

Table 4: tiefpass (raw data)